Kepler and Newton as Machine Learners: From Tycho Brahe's Data to Universal Gravitation

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Abstract

Between 1600 and 1609, Johannes Kepler performed history's first large-scale nonlinear least-squares regression: fitting ellipse parameters to Tycho Brahe's Mars observations, which achieved then-unprecedented $2'$-arcminute precision. Eighty years later, Isaac Newton derived the law of universal gravitation from Kepler's three empirical laws by a sequence of deductions that constitute a hypothetico-deductive inference chain. This paper examines the two-century episode from the perspective of modern statistics and machine learning. Kepler's problem was nonlinear parameter estimation: the elliptical model $r(\theta) = a(1 - e^2)/(1 + e\cos\theta)$ was new, the parameters (a, e) unknown, and the loss function the sum of squared angular residuals. The optimal Mars parameters — $a = 1.524\,\mathrm{AU}$, $e = 0.0934$ — can be recovered in seconds by nonlinear least squares or MCMC; what required Kepler nearly a decade was the prior discovery of the correct functional form. The 8-arcminute residual of the circular hypothesis played the role of a four-sigma outlier: statistically inconsistent with Brahe's measurement precision, it forced a model expansion from circular to elliptical orbits. Kepler's Third Law $T^2 \propto a^3$, which he discovered inductively across the five planets then known, is a log-log OLS regression: the slope estimate is $\hat{\beta} = 1.500$ and $R^2 = 0.99998$. Newton's contribution was an inverse problem: given Kepler's three empirical regularities as constraints, identify the unique force law $F(r)$ that produces them. The answer, $F \propto r^{-2}$, follows from elementary calculus and the Binet equation; the Moon test then confirmed out-of-sample generalization from planetary to terrestrial gravitation. Modern AI systems — *AI Feynman* [Udrescu and Tegmark, 2020], *AI-Newton* [Fang et al., 2023], and the deep-learning network

of Lemos et al. [2023] — reconstruct the same laws from trajectory data with no prior physics knowledge. The contrast illuminates both the power of theory-informed inference and the irreducible role of conceptual insight in scientific discovery.

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1 Introduction

A modern data scientist confronted with a table of (orbital period, semi-major axis) pairs would recognise Kepler's Third Law as a two-variable power-law regression, estimated in about one line of Python. A data scientist given a time series of planetary positions

would frame the orbit-fitting problem as nonlinear least squares with two free parameters. Neither would need Kepler's conceptual leap —the abandonment of circles in favour of ellipses —because they would already know the functional form. What Kepler *did* need, and what took him nine years and hundreds of pages of *Astronomia Nova* [Donahue, 1992], was to discover the model class before estimating its parameters.

The episode repays examination from a statistical perspective for three reasons.

First, the *data*. Tycho Brahe's naked-eye observations of Mars, accumulated over twenty years at Uraniborg, achieved $2'$ (arcminute) precision —roughly ten times better than anything available to Copernicus — and were thus sufficient to distinguish between competing orbit models. Brahe's archive was the training set; its precision determined the statistical power available to any estimation procedure.

Second, the *model-selection criterion*. Kepler's pivotal insight was not a computation but a residual diagnostic: when the Vicarious Hypothesis (an eccentric circle fitted to Brahe's data) predicted Mars longitudes within $2'$ but left an irreducible $8'$ error at certain orbital phases, Kepler recognised that $8' = 4\sigma$ was statistically incompatible with measurement noise. He could not ignore it, and so was forced to enlarge the model class. The ellipse —the next simplest closed curve after the circle —fitted the residuals to within the noise floor, and no further expansion was needed. This is textbook model selection: reject the null model when the residual exceeds a critical threshold, expand the hypothesis class, and stop as soon as the residuals are consistent with noise.

Third, the *identification problem*. Newton's contribution was not more data or a better fitting algorithm. It was the recognition that Kepler's three empirical laws, taken together, over-identify a unique force law. The area law (equal areas in equal times) implied a central force. The harmonic law ($T^2 \propto a^3$) implied an inverse-square dependence on distance. The fact that orbits are ellipses — not parabolas or other conics — confirmed the inverse-square law by the converse of Proposition 11, Book I of the *Principia*. The Moon test extended the law from the solar system to the Earth–Moon system, establishing universality across seven orders of magnitude in distance.

Seen through a machine-learning lens, Brahe was the data engineer, Kepler was the model builder who performed nonlinear estimation under a parsimony constraint, and Newton solved the inverse problem of recovering the latent physical law from Kepler's compressed representation of the data. Modern AI systems have now automated all

three steps. The remainder of this paper develops each step in detail, provides numerical worked examples, and closes with a comparison of Kepler and Newton's reasoning with the methods of contemporary machine learning.

2 Tycho Brahe's Data and the Mars Problem

2.1 Observational Precision

Tycho Brahe (1546–1601) constructed the most precise pre-telescopic observational instruments in history at his observatory Uraniborg on the island of Hven. His quadrants and sextants measured celestial angles to approximately $2'$ (arcminutes), an order of magnitude more precisely than his predecessors. Over two decades he accumulated positions for all of the then-known planets, the Moon, and over a thousand fixed stars.

Two arcminutes is a small angle —$(2/60) \times (\pi/180) = 5.8 \times 10^{-4}$ radians —but it is large enough to matter. At Mars's mean distance of 1.524 AU from the Sun, a $2'$ angular error corresponds to a linear uncertainty of

$$\delta r = 1.524 \times 5.8 \times 10^{-4}\,\text{AU} = 8.8 \times 10^{-4}\,\text{AU} \approx 0.001\,\text{AU},$$

or about 130,000 km. This is the irreducible noise floor in Kepler's estimation problem.

2.2 The Method of Oppositions

Kepler used Mars observations taken at *opposition* — moments when the Sun, Earth, and Mars are collinear, with Earth in the middle. At opposition, the geocentric and heliocentric longitudes of Mars coincide (because the Earth–Sun–Mars angle is zero), so no geometric correction for the Earth's position is needed. Kepler identified twelve usable oppositions in Brahe's archive spanning 1580 to 1600. Each opposition supplied one data point (r_i, θ_i): the heliocentric distance r_i and the heliocentric true anomaly θ_i of Mars.

2.3 Table 1: Selected Brahe–Kepler Opposition Data for Mars

Table 1 lists five of Kepler's Mars opposition data points in the form convenient for parameter estimation. The values are approximate but historically representative, with

added noise consistent with the $2'$ precision of Brahe's instruments.

Table 1: Selected Mars opposition heliocentric coordinates derived from Brahe's observations. Distances in AU; true anomaly θ measured from perihelion. Noise is of order 0.001–0.002 AU, consistent with Brahe's $2'$ angular precision at 1.5 AU distance.

i	θ_i (degrees)	r_i (AU)
1	0	1.382
2	75	1.477
3	150	1.641
4	225	1.619
5	300	1.442

3 Kepler's Parametric Model

3.1 The Vicarious Hypothesis and Its Failure

Kepler's initial working model for Mars was the *Vicarious Hypothesis*: an eccentric circle, with an equant point offset from the centre of the circle. This model had been the standard since Ptolemy. Fitted to oppositions, it predicted heliocentric longitudes within $2'$ — inside Brahe's error budget —and was thus statistically acceptable on longitude residuals alone.

The problem appeared when Kepler checked the *latitudes* and distances. At orbital phases halfway between opposition and conjunction, the circular model produced errors of $8'$ —four times Brahe's precision. Kepler wrote in *Astronomia Nova* (Chapter 19):

> *"If thou, dear reader, art bored with this wearisome method of calculation, take pity on me who had to go through with at least seventy repetitions of it... But if you are wearied by this prolixity, you will be still more wearied when I tell you that this hypothesis is nevertheless wide of the truth."*

And, crucially (Chapter 19):

> *"These eight minutes alone will have led the way to the reformation of all of astronomy."*

3.2 The Ellipse

After confirming that no circular hypothesis could explain Brahe's data, Kepler tried the oval and ultimately the ellipse. The ellipse, in polar coordinates centred at one focus (the Sun), is

$$\boxed{r(\theta) = \frac{a(1-e^2)}{1+e\cos\theta}} \tag{1}$$

where a is the semi-major axis, $e \in (0,1)$ is the eccentricity, and θ is the true anomaly measured from perihelion. Two free parameters, (a, e), specify the size and shape of the orbit. A third parameter, the argument of perihelion, specifies the orientation of the ellipse in space, but this can be absorbed into the definition of $\theta = 0$.

3.3 Kepler's Equation

The ellipse specifies the orbit's *shape*: where the planet is at a given true anomaly. To predict the planet's position at a given *time*, one needs to connect θ to time via the eccentric anomaly E and the mean anomaly M. The key relation is *Kepler's equation*:

$$\boxed{M = E - e\sin E} \tag{2}$$

where $M = 2\pi(t-t_0)/T$ increases linearly with time and E is related to θ by $\tan(\theta/2) = \sqrt{(1+e)/(1-e)}\,\tan(E/2)$. Kepler's equation (2) is transcendental in E and must be solved iteratively, an additional nonlinearity that Kepler handled by successive approximation —a procedure equivalent to Newton's method applied to $g(E) = M - E + e\sin E = 0$.

4 The Eight-Minute Pivot: Model Selection by Residual Diagnostics

4.1 The Circular Model as a Null Hypothesis

The Vicarious Hypothesis —optimally fitted to Brahe's data —predicted Mars longitudes within $2'$ at opposition epochs, consistent with measurement noise. The hypothesis could not be rejected on longitude residuals.

The problem was that it predicted Mars distances incorrectly. Converting distance

errors to angular terms at Mars's 1.524 AU distance, the deviation reached $8'$. Since Brahe's measurement precision was $\sigma \approx 2'$, the circular model's maximum residual was

$$t_{\max} = \frac{8'}{2'} = 4 \text{ standard deviations.}$$

Under a Gaussian noise model, a four-sigma deviation occurs with probability $p = 2\Phi(-4) \approx 6.3 \times 10^{-5}$. The circular hypothesis was decisively rejected.

4.2 Model Expansion

The rejection of the circular null drove Kepler to enlarge his model class in the direction of greater flexibility. The sequence of candidates, ordered by complexity, was:

1. Circle (one shape parameter: radius a).
2. Eccentric circle with equant (two geometric parameters: eccentricity e, equant offset d). This is the Vicarious Hypothesis.
3. Oval (one additional shape parameter).
4. Ellipse (two parameters: a, e).

Kepler tried the oval and found it over-corrected at some phases while under-correcting at others. The ellipse, with its elegant two-parameter form, reduced all residuals to within $2'$ —inside the noise floor — and passed the residual test. No richer model was needed.

This is the logic of the *Minimum Description Length* (MDL) principle [Udrescu and Tegmark, 2020, Schmidt and Lipson, 2009]: prefer the model that minimises the sum of (i) the cost of encoding the model's parameters and (ii) the cost of encoding the residuals under the model. The ellipse minimises this sum because it captures the systematic structure in Brahe's data with only two parameters, leaving only random noise in the residuals.

5 Nonlinear Least Squares for the Ellipse Parameters

5.1 The NLS Objective

Given n opposition observations $\{(\theta_i, r_i)\}_{i=1}^n$ and an assumed noise model $r_i = f(\theta_i; a, e) + \varepsilon_i$ where $\varepsilon_i \overset{iid}{\sim} (0, \sigma^2)$, the NLS estimator minimises the sum of squared residuals:

$$(\hat{a}, \hat{e}) = \arg\min_{a,e} \sum_{i=1}^{n} \big[r_i - f(\theta_i; a, e)\big]^2 \tag{3}$$

where $f(\theta; a, e) = a(1 - e^2)/(1 + e\cos\theta)$, as in equation (1).

Unlike OLS, the NLS problem (3) is nonlinear in the parameters. No closed-form solution exists. The standard algorithm is *Gauss–Newton*: starting from an initial guess $(a^{(0)}, e^{(0)})$, each iteration solves a linearised least-squares problem. Let $\mathbf{J}^{(k)}$ be the $n \times 2$ Jacobian matrix with

$$J_{i1}^{(k)} = \frac{\partial f}{\partial a}\Big|_{\theta_i,\, a^{(k)}, e^{(k)}} = \frac{1 - e^2}{1 + e\cos\theta_i}, \qquad J_{i2}^{(k)} = \frac{\partial f}{\partial e}\Big|_{\theta_i,\, a^{(k)}, e^{(k)}} = \frac{-a(2e + (1 + e^2)\cos\theta_i)}{(1 + e\cos\theta_i)^2}.$$

The Gauss–Newton update is

$$\begin{pmatrix} a^{(k+1)} \\ e^{(k+1)} \end{pmatrix} = \begin{pmatrix} a^{(k)} \\ e^{(k)} \end{pmatrix} + \big((\mathbf{J}^{(k)})^\top \mathbf{J}^{(k)}\big)^{-1} (\mathbf{J}^{(k)})^\top \mathbf{r}^{(k)}, \tag{4}$$

where $\mathbf{r}^{(k)}$ is the residual vector at the k-th iterate. The algorithm converges in a few iterations when started near the solution. Kepler's iterative procedure —adjusting a and e by hand, checking residuals, and revising —is logically equivalent to early iterations of Gauss–Newton, with human judgement substituting for the matrix product.

5.2 NLS Residual Table for Mars

Table 2 shows the NLS fit evaluated at the converged estimates $\hat{a} = 1.524\,\text{AU}$ and $\hat{e} = 0.0934$, applied to the five observations in Table 1.

Table 2: NLS residuals for the five Mars opposition observations. Fitted model: $\hat{r}(\theta) = \hat{a}(1-\hat{e}^2)/(1+\hat{e}\cos\theta)$ with $\hat{a} = 1.524\,\text{AU}$, $\hat{e} = 0.0934$. All residuals lie within Brahe's $2'$ precision ($\approx 0.002\,\text{AU}$).

i	θ_i	r_i (AU)	$\hat{r}_i$ (AU)	$\hat{\varepsilon}_i$ (AU)
1	$0°$	1.382	1.381	$+0.001$
2	$75°$	1.477	1.475	$+0.002$
3	$150°$	1.641	1.643	-0.002
4	$225°$	1.619	1.618	$+0.001$
5	$300°$	1.442	1.443	-0.001
$SS_{\text{res}} = \sum \hat{\varepsilon}_i^2$				$1.1 \times 10^{-5}\,\text{AU}^2$
$\text{RMSE} = \sqrt{SS_{\text{res}}/n}$				$0.00148\,\text{AU}$

The RMSE of $0.00148\,\text{AU}$ corresponds to an angular error of approximately

$$\frac{0.00148}{1.524} \times \frac{180°}{\pi} \times 60 \approx 3.3'$$

at Mars's mean distance. This lies inside Brahe's observational noise floor of $2'$–$3'$, confirming that the ellipse with two parameters absorbs all the systematic structure in the data.

5.3 Historical Significance of the Residual Reduction

The eight-minute rejection of Section 4 and the NLS convergence of this section are two sides of the same statistical coin. Before the ellipse: RMSE $\approx 4'$, four times the noise floor. After the ellipse: RMSE $\approx 2'$–$3'$, inside the noise floor. The improvement in SS_{res} from the circular to the elliptical model is a factor of roughly $4^2 = 16$, at the cost of enlarging the parameter space from one (the radius a) to two (a, e). Any model-selection criterion —adjusted R^2, AIC, BIC, MDL —awards this trade-off overwhelmingly to the ellipse.

6 Maximum Likelihood and the Cramér–Rao Bound

6.1 The Gaussian Noise Model

Assuming independent Gaussian measurement errors, $r_i = f(\theta_i; a, e) + \varepsilon_i$ with $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$, the log-likelihood is

$$\ell(a, e, \sigma^2 \mid \mathbf{r}) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\big[r_i - f(\theta_i; a, e)\big]^2.$$

Maximising over (a, e) for fixed σ^2 is equivalent to minimising $\sum[r_i - f(\theta_i; a, e)]^2$, i.e. the NLS objective (3).

Theorem (MLE = NLS). *Under independent Gaussian observation errors, the maximum likelihood estimator* $(\hat{a}^{MLE}, \hat{e}^{MLE})$ *coincides with the NLS estimator* $(\hat{a}, \hat{e})$.

The proof is immediate from the log-likelihood above: the Gaussian distributional assumption enters only through the quadratic penalty, and the maximiser over (a, e) is therefore the same as the minimiser of the sum of squared residuals.

6.2 Fisher Information and the Cramér–Rao Bound

Let $\boldsymbol{\phi} = (a, e)^\top$ denote the parameter vector. The Fisher information matrix is

$$\mathcal{I}(\boldsymbol{\phi}) = \frac{1}{\sigma^2}\mathbf{J}^\top\mathbf{J},$$

where $\mathbf{J}$ is the $n \times 2$ Jacobian evaluated at the true parameters (i.e. $\mathbf{J}$ has i-th row $[\partial f/\partial a,\ \partial f/\partial e]$ evaluated at θ_i and the true (a, e)). The Cramér–Rao bound states that for any unbiased estimator $\tilde{\boldsymbol{\phi}}$,

$$\mathrm{Var}(\tilde{\boldsymbol{\phi}}) \ \geq\ \mathcal{I}(\boldsymbol{\phi})^{-1} = \sigma^2(\mathbf{J}^\top\mathbf{J})^{-1}, \tag{5}$$

where the inequality holds matrix-element-wise. The NLS/MLE estimator achieves this bound asymptotically.

6.3 Design Implications: Coverage and Variance Reduction

The matrix $\mathbf{J}^\top\mathbf{J}$ depends on the distribution of the observation angles $\{\theta_i\}$. The partial derivatives are:

$$\frac{\partial f}{\partial a} = \frac{1-e^2}{1+e\cos\theta}, \qquad \frac{\partial f}{\partial e} = -\frac{a(2e+(1+e^2)\cos\theta)}{(1+e\cos\theta)^2}.$$

Because $\partial f/\partial e$ depends on $\cos\theta$, observations near perihelion ($\theta = 0$) and aphelion ($\theta = 180°$) provide maximum information about e: the planet's radial excursion from a circle is largest near the apsides. Observations bunched near $\theta = 90°$ or $\theta = 270°$ contribute less to eccentricity estimation because the ellipse and a circle of the right radius agree closely near the semi-latus rectum.

Table 3 summarises the design implications of bound (5).

Table 3: Design implications of the Cramér–Rao bound for NLS estimation of (a, e). The bound favours observations spread across all orbital longitudes, with particular emphasis on the apsidal directions $\theta \approx 0°$ and $\theta \approx 180°$.

Design choice	**Effect on $\mathbf{J}^\top\mathbf{J}$**	**Effect on variance**
More observations (large n)	All eigenvalues of $\mathbf{J}^\top\mathbf{J}$ increase	Both $\mathrm{Var}(\hat{a})$ and $\mathrm{Var}(\hat{e})$ fall as $O(1/n)$
Wider angular coverage (all θ)	Off-diagonal term $\sum \partial f/\partial a \cdot \partial f/\partial e$ cancels; matrix becomes more diagonal	Parameters become better separated; covariance falls
Apsidal observations ($\theta \approx 0°$ or $180°$)	$\lvert\partial f/\partial e\rvert$ is maximised, increasing the $(2,2)$ element	$\mathrm{Var}(\hat{e})$ is minimised
Higher instrument precision (small σ^2)	$\mathcal{I} = \sigma^{-2}\mathbf{J}^\top\mathbf{J}$ inflates	Both variances fall proportionally

Brahe's twenty years of observations, spread across two orbital periods of Mars, provided many oppositions at a wide range of true anomalies — exactly the design that minimises the Cramér–Rao bound. Kepler's own comment on the precision of Brahe's archive — "Even if the positions of Mars were off by two or three minutes I would be unable to say whether the orbit was an oval or an ellipse" — is a practical statement about the bound: without sub-2′ precision, the eccentricity $e = 0.0934$ would not be identifiable

against a circular hypothesis.

7 Kepler's Third Law: A Log-Log Regression

7.1 The Empirical Discovery

In *Harmonices Mundi* (1619), Kepler reported a relation he had sought for over a decade: the square of a planet's orbital period is proportional to the cube of its semi-major axis,

$$\boxed{T^2 = k\,a^3} \tag{6}$$

with k a universal constant (the same for all planets orbiting the Sun). Kepler found this law by scanning tables of T^2 and a^3 for the five planets then known until the constant ratio T^2/a^3 became apparent.

7.2 Log-Log OLS

Taking logarithms of (6) gives

$$\log T = \frac{3}{2}\log a + \tfrac{1}{2}\log k.$$

This is a bivariate linear regression $Y_i = \beta X_i + \alpha + \varepsilon_i$ with $Y_i = \log T_i$, $X_i = \log a_i$, $\beta = 3/2$, and $\alpha = \frac{1}{2}\log k$. OLS gives

$$\hat{\beta} = \frac{S_{XY}}{S_{XX}}, \qquad \hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X},$$

where $S_{XY} = \sum(X_i - \bar{X})(Y_i - \bar{Y})$ and $S_{XX} = \sum(X_i - \bar{X})^2$.

7.3 Table 3: Kepler's Third Law for the Six Known Planets

Table 4 assembles the data for Mercury through Saturn (the six planets available to Kepler and his contemporaries) in AU and Julian years.

Table 4: Orbital data for the six planets known in Kepler's time. Periods T in years, semi-major axes a in AU (1 AU = Earth's mean distance from the Sun). The ratio T^2/a^3 is 1.000 ± 0.001 for all six planets. Log values are natural logarithms.

Planet	a (AU)	T (yr)	T^2	a^3	T^2/a^3	$\ln a$	$\ln T$
Mercury	0.387	0.241	0.0581	0.0580	1.001	−0.949	−1.423
Venus	0.723	0.615	0.378	0.378	1.000	−0.324	−0.486
Earth	1.000	1.000	1.000	1.000	1.000	0.000	0.000
Mars	1.524	1.881	3.538	3.540	0.999	0.421	0.632
Jupiter	5.203	11.86	140.7	140.8	0.999	1.649	2.473
Saturn	9.537	29.46	867.5	867.1	1.001	2.255	3.383

7.4 OLS Computation

With $n = 6$, $\bar{X} = 0.509$, $\bar{Y} = 0.763$:

$$S_{XX} = \sum_{i=1}^{6}(X_i - \bar{X})^2 = 7.431, \qquad S_{XY} = \sum_{i=1}^{6}(X_i - \bar{X})(Y_i - \bar{Y}) = 11.147.$$

Hence

$$\hat{\beta} = \frac{11.147}{7.431} = 1.500, \qquad \hat{\alpha} = 0.763 - 1.500 \times 0.509 = -0.001 \approx 0.$$

The slope estimate is $\hat{\beta} = 1.500$, consistent to four significant figures with the theoretical value of 3/2. The intercept estimate $\hat{\alpha} \approx 0$ implies $k \approx e^0 = 1.000$ in natural units of yr^2/AU^3.

The OLS residuals $\hat{\varepsilon}_i = Y_i - \hat{Y}_i$ are: Mercury: -0.0004; Venus: $+0.0006$; Earth: $+0.0001$; Mars: 0.0000; Jupiter: $+0.0004$; Saturn: -0.0002. The residual sum of squares is $SS_{\text{res}} = 7.3 \times 10^{-7}$, giving

$$R^2 = 1 - \frac{SS_{\text{res}}}{S_{YY}} = 1 - 4.4 \times 10^{-8} \approx 0.99998.$$

With $n-2 = 4$ degrees of freedom, the standard error of the slope is $\text{SE}(\hat{\beta}) = \hat{\sigma}/\sqrt{S_{XX}} = 0.000428/2.726 = 0.000157$, giving a t-statistic of $\hat{\beta}/\text{SE}(\hat{\beta}) = 9{,}554$.

7.5 Identification: What Does k Measure?

The OLS regression identifies the slope $\hat{\beta} = 3/2$ and the intercept $\hat{\alpha} \approx 0$, but it does not by itself identify the physical constant k. For that, Newton's mechanics is required. Newton showed that for a central $F \propto 1/r^2$ force of magnitude $F = GM_S m/r^2$, a circular orbit at radius a satisfies

$$\frac{4\pi^2 a}{T^2} = \frac{GM_S}{a^2} \implies T^2 = \frac{4\pi^2}{GM_S}\, a^3.$$

Thus $k = 4\pi^2/(GM_S)$, where G is Newton's gravitational constant and M_S the Sun's mass. In AU–yr units, $k = 1.000\,\mathrm{yr}^2/\mathrm{AU}^3$, which implies $GM_S = 4\pi^2 \approx 39.48\,\mathrm{AU}^3/\mathrm{yr}^2$.

This structure is analogous to the measurement-equation identification problem in the photoelectric effect [see Casella and Berger, 2002, Chapter 7]: OLS identifies a *ratio* of physical constants (h/e in the photoelectric case, GM_S in the orbital case), and a second measurement is needed to extract each constant separately.

8 Newton's Inverse Problem and the Force Law

8.1 Overview of the Inference Chain

Newton's achievement in the *Principia* [Newton, 1687] was to show that Kepler's three empirical laws are theorems, not accidents. Starting from three observed regularities — the area law, the elliptical orbit shape, and the harmonic law —Newton identified the unique force law consistent with all three simultaneously. This is an *inverse problem*: given outputs (Kepler's laws), recover the latent mechanism (the force law). Table 5 traces the four-step logical chain.

8.2 The Binet Equation

The key mathematical result in Step C is the Binet equation, derived from Newton's second law in polar coordinates ($F_r = m(\ddot{r} - r\dot{\theta}^2)$) via the substitution $u = 1/r$:

$$\frac{d^2u}{d\theta^2} + u = -\frac{F(1/u)}{mh^2u^2}. \tag{7}$$

For an inverse-square force $F = -GMmu^2$ (attractive), the right-hand side is the constant GM/h^2, giving the linear second-order ODE

$$\frac{d^2u}{d\theta^2} + u = \frac{GM}{h^2},$$

whose general solution is

$$u(\theta) = \frac{GM}{h^2}\big[1 + e\cos(\theta - \theta_0)\big], \qquad \text{i.e.,} \quad r(\theta) = \frac{p}{1 + e\cos\theta}, \quad p = \frac{h^2}{GM}.$$

This is equation (1) with $p = a(1 - e^2)$. Newton's genius was to run the Binet argument in reverse: *given* that orbits are ellipses (Kepler's First Law), the Binet equation implies the right-hand side is constant, hence $F \propto u^2 = r^{-2}$.

9 The Moon Test: Out-of-Sample Generalization

9.1 The Numerical Argument

The Moon test is Newton's out-of-sample validation of the inverse-square law. Armed with Picard's (1671) improved measurement of Earth's radius [Picard, 1671] ($R_\oplus = 6{,}371\,\text{km}$), Newton computed the Moon's centripetal acceleration from its measured orbital period and radius, and compared it to the surface gravitational acceleration.

The Moon orbits at $r_M = 384{,}400\,\text{km} = 60.3\,R_\oplus$ with period $T_M = 27.3\,\text{days} = 2.36 \times 10^6\,\text{s}$. Its centripetal acceleration is

$$a_M = \frac{4\pi^2 r_M}{T_M^2} = \frac{4\pi^2 \times 3.844 \times 10^8}{(2.36 \times 10^6)^2} = 0.00272\,\text{m/s}^2. \tag{8}$$

Under the inverse-square law, the ratio of surface gravity to lunar centripetal acceleration should equal the square of the distance ratio:

$$\frac{g}{a_M} \stackrel{?}{=} \left(\frac{r_M}{R_\oplus}\right)^2 = (60.3)^2 = 3{,}636.$$

With $g = 9.80\,\text{m/s}^2$ and $a_M = 0.00272\,\text{m/s}^2$:

$$\left.\frac{g}{a_M}\right|_{\text{observed}} = \frac{9.80}{0.00272} = 3{,}603.$$

The observed ratio 3,603 exceeds the predicted 3,636 by $(3{,}636 - 3{,}603)/3{,}636 = 0.9\%$ — agreement well within the measurement uncertainty in Earth's radius and the Moon's orbital data available to Newton.

9.2 Significance of the Moon Test

The Moon test is the first *generalization* test in the history of physics. Newton's model was calibrated to solar-system data (Kepler's laws); the Moon test applied it to a completely different system (Earth–Moon) at seven orders of magnitude closer range. In machine-learning terms, it is an out-of-sample evaluation on a held-out test set drawn from a different distribution. The fact that the model passed —with an error under 1% —provided compelling evidence for the universality of the inverse-square law.

10 AI Reconstructions of Kepler's Laws and Newton's Gravity

10.1 AI Feynman and Symbolic Regression

Symbolic regression (SR) searches the space of mathematical expressions — combinations of arithmetic operations, trigonometric functions, and powers — for the formula that best fits a data set subject to a parsimony penalty. *AI Feynman* [Udrescu and Tegmark, 2020] extends standard SR with physics-inspired simplifications: it exploits dimensional analysis, decomposition into simpler sub-functions, and neural pre-smoothing of noisy data before the symbolic search begins.

Fed a data set of (r, θ) pairs generated from planetary orbits, AI Feynman recovers

$$r = \frac{p}{1 + e\cos\theta}$$

as the most parsimonious description, landing at the "kink" in the Pareto frontier (accuracy vs. complexity) where the ellipse achieves a large accuracy gain at modest complexity cost.

10.2 AI-Newton

The *AI-Newton* system of Fang et al. [2023] is concept-driven: it works with raw multi-experiment data, identifies physical concepts (e.g. "central force", "conserved angular momentum"), and generalises them into symbolic laws. Given planetary trajectory data —without being told Kepler's laws or Newton's —it rediscovered:

- central-force behaviour (Step A in Table 5),
- the inverse-square force law (Step B),
- Newton's second law and conservation of energy.

The conceptual progression —trajectories to central force to inverse-square law —mirrors Newton's own chain, executed without human supervision.

10.3 Deep Learning: Lemos et al. (2023)

Lemos et al. [2023] trained a graph neural network on approximately thirty years of Solar System trajectory data. The network architecture encodes the inductive bias that interactions are pairwise and radial. After training, symbolic regression was applied to the learned force representation and recovered $F \propto r^{-2}$, matching Newton's law. The network also inferred the relative masses of the planets from trajectory curvatures —masses are not observed directly but enter as latent variables in the Newtonian model.

10.4 Pareto Frontier and MDL

A useful organising principle for comparing models of different complexity is the *Pareto frontier* in the accuracy–complexity plane. As one moves from circular to elliptical to higher-order models, accuracy improves but complexity grows. The ellipse sits at the dominant kink in the frontier: beyond it, complexity grows steeply while accuracy improves only marginally. This kink is the hallmark of a true physical law as opposed to an over-fitted curve.

The information-theoretic analogue is the Minimum Description Length (MDL) principle [Schmidt and Lipson, 2009]: a good model is one that compresses the data. The MDL

cost of a model is

$$L(\text{Model}) = L(\text{Hypothesis}) + L(\text{Data} \mid \text{Hypothesis}),$$

where $L(\cdot)$ is code length in nats. The ellipse achieves a far shorter total description length than any polynomial or Fourier series of comparable accuracy, because its two-parameter structure encodes the physical geometry rather than merely fitting the data.

10.5 Neural Smoothing

Brahe's $2'$ observational noise adds jitter to the (r, θ) data. Modern AI Feynman [Udrescu and Tegmark, 2020] handles this by first training a neural network to interpolate the noisy observations, then querying the smoothed neural manifold for the symbolic search. This "neural denoising" step allows SR to focus on the signal rather than the noise — exactly as Kepler discarded the small ($< 2'$) residuals of the ellipse and focused on the systematic $8'$ excess of the circular model.

11 Kepler and Newton as Machine Learners

11.1 The Shared Structure

Both Kepler and Newton proceeded in a pattern that is now standard in machine learning: they identified a hypothesis class, specified features, chose a loss function, fitted the model to data, and validated out of sample. What was historically novel in each step reflects the limits of their era, not any departure from the ML framework.

11.2 Hypothesis Class and Inductive Bias

Kepler's *implicit hypothesis class* was the family of smooth closed curves: first circles, then ovals, then conic sections. His *inductive bias* —the preference embedded in his search strategy — was for geometric parsimony. He accepted the ellipse not because he had exhausted all alternatives but because it was the simplest curve that explained the data.

Newton's hypothesis class was more structured: the family of central force laws $F = F(r)$,

parametrised by the function $F(\cdot)$. His inductive bias was the requirement that *all three* of Kepler's laws be jointly satisfied. This multi-constraint identification reduced the hypothesis class to a single function: $F \propto r^{-2}$.

11.3 Feature Engineering

Kepler's key feature-engineering decision was the choice of coordinates. Ptolemaic astronomy used geocentric Cartesian coordinates; Copernicus shifted to heliocentric Cartesian. Kepler's decisive step was to use heliocentric *polar* coordinates (r, θ) with the Sun at the focus. In this representation, the ellipse has a simple two-parameter closed form (equation (1)). In Cartesian coordinates, the same ellipse requires five parameters ($x^2/a^2 + y^2/b^2 = 1$ plus four transformation parameters for position and orientation). Choosing the right representation halved the parameter space and made the nonlinearity manageable —a classic feature-engineering gain.

11.4 Loss Function and Model Selection

Kepler's effective loss function was the maximum angular residual, not the mean squared residual. He rejected the circular model when its maximum prediction error ($8'$) exceeded Brahe's precision ($2'$) by a factor of four. This is analogous to a max-norm or L_∞ loss, which is appropriate when the measurement noise has a known hard upper bound. In modern terms, Kepler performed a model-selection step based on whether the worst-case residual was consistent with the noise model —a robust form of hypothesis testing.

11.5 Generalisation and Transfer Learning

Newton's extension of the inverse-square law from the solar system to the Earth–Moon system is *transfer learning*: a model learnt in one domain (heliocentric planetary orbits) is applied without re-training to a different domain (geocentric lunar orbit). The Moon test quantified the generalisation gap and found it to be less than 1%. Newton then extended the same law to the tides, comets, and eventually to falling bodies near Earth's surface —a series of zero-shot applications across domains that differ by many orders of magnitude in mass, distance, and velocity.

11.6 The Hypothetico-Deductive Loop

Table 6 organises the full Brahe–Kepler–Newton episode into the hypothetico-deductive (H-D) loop that is common to both scientific inference and modern machine learning.

11.7 What ML Cannot Replicate

Modern AI systems can rediscover Kepler's orbit law and Newton's force law from trajectory data (Section 10). What they have not yet automated is the *conceptual* leap that connects the two.

Kepler needed a new *physical concept* —that the Sun exerts a guiding force on the planets —to motivate the hypothesis that orbits close around the Sun at the focus of the ellipse. Without that concept, no computable loss function points uniquely toward the focus form (1).

Newton needed the concept of *inertia* to separate force from motion: equal areas in equal times is not intuitively a statement about force until one has Newton's first law. The Binet equation is a relation between the force law and the orbit shape, but writing it down required the concept of a continuously acting force that curves what would otherwise be straight-line motion.

Current symbolic-regression systems recover the formulas but not these bridging concepts. *AI-Newton* [Fang et al., 2023] is a partial exception —it identifies "central force" as an intermediate concept —but it does so by hard-coding a library of physical primitives. A fully autonomous scientific agent would need to construct those primitives itself, which remains an open problem.

12 Comparison of Methods

Table 7 locates Kepler's, Newton's, and modern methods on a common set of dimensions.

13 Conclusions

The Brahe–Kepler–Newton episode is one of the most instructive examples of scientific inference in history. Examined through the lens of modern statistics and machine learning, it exhibits a coherent pipeline:

1. **Data collection** (Brahe): twenty years of $2'$-precision Mars observations, sufficient to distinguish an ellipse ($e = 0.0934$) from a circle ($e = 0$) at the 4σ level.
2. **Model selection by residual diagnostics** (Kepler): the $8'$ systematic error of the circular model exceeded the noise floor by a factor of four, forcing model expansion. The ellipse absorbed the excess with only two parameters, confirming it as the Pareto-optimal fit.
3. **Nonlinear least squares** (Kepler): the ellipse parameters $a = 1.524\,\mathrm{AU}$ and $e = 0.0934$ were converged upon by approximately seventy iterative adjustments — logically equivalent to Gauss–Newton iterations.
4. **Log-log OLS** (Kepler): the harmonic law $T^2 \propto a^3$ is a one-slope log-log regression with $\hat{\beta} = 1.500$, $\hat{\alpha} \approx 0$, and $R^2 = 0.99998$ over six planets — a cleaner fit than almost any empirical relationship in the physical sciences.
5. **Inverse problem and identification** (Newton): Kepler's three laws jointly identify the inverse-square force law $F \propto r^{-2}$ uniquely. The Binet equation formalises Newton's geometric deduction in modern calculus.
6. **Out-of-sample generalization** (Newton): the Moon test confirmed the inverse-square law at $r = 60.3\,R_\oplus$ with a 0.9% discrepancy —an out-of-sample validation across seven orders of magnitude in distance.

Modern AI systems reproduce all six steps automatically. *AI Feynman* recovers the ellipse from (r, θ) data. The Lemos et al. network recovers $F \propto r^{-2}$ from trajectory data. *AI-Newton* reproduces the conceptual chain from trajectories to central force to inverse-square law without prior physics knowledge.

What the AI systems have not yet fully automated is the *conceptual architecture* that makes the problem tractable: the heliocentric polar coordinate representation, the identification of the Sun as the force centre, and the Newtonian concept of inertia that distinguishes "force" from "sustained velocity." Kepler and Newton were machine learners in a deep sense, but they were also concept engineers —and modern AI has only partially closed that gap.

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Table 5: Newton's four-step deductive chain connecting Kepler's empirical laws to the inverse-square gravitational force. Each step is stated in Newton's geometric language and in modern calculus.

Step	**Input**	**Newton's argument**	**Modern calculus**
A	Kepler's Second Law: equal areas in equal times	Areas swept by the radius vector are equal in equal times. Newton shows (Prop. 1, Book I) this is equivalent to zero tangential force component.	$h = r^2\dot{\theta} = \text{const}$; hence $\ddot{r} - r\dot{\theta}^2 = F_r$ and $F_\theta = 0$: the force is central.
B	Kepler's Third Law: $T^2 \propto a^3$	For a circular orbit, centripetal acceleration is $4\pi^2 r/T^2$. Substituting $T^2 = ka^3$ and $a = r$: acceleration $\propto r^{-2}$.	$F = m \cdot \frac{4\pi^2}{k} \cdot r^{-2}$; identifying $k = 4\pi^2/(GM)$ gives $F = GMm/r^2$.
C	Force is central and $\propto r^{-2}$	Prop. 11, Book I: conic sections are produced by an inverse-square central force. Conversely, an inverse-square force implies conic orbit (Prop. 13).	Binet equation: $u'' + u = GM/h^2$ (where $u = 1/r$); solution $u \propto 1 + e\cos\theta$, i.e. $r = p/(1 + e\cos\theta)$: an ellipse.
D	Inverse-square law from solar orbits	Moon test: at Earth's surface, $g = 9.8\,\text{m/s}^2$. At Moon's distance $r_M = 60.3\,R_\oplus$, predicted centripetal acceleration $g/(60.3)^2 = 0.00269\,\text{m/s}^2$. Observed: $0.00272\,\text{m/s}^2$ (agreement 0.9%).	$g_{\text{Moon}} = GM_\oplus r_M^{-2} = g_{\text{surf}}(R_\oplus/r_M)^2$; universality confirmed.

Table 6: The Brahe–Kepler–Newton episode as a hypothetico-deductive (H-D) loop. Each row maps a historical step to a corresponding concept in modern machine learning.

H-D step	**Historical action**	**ML analogue**
Data collection	Brahe measures Mars positions over 20 years at $2'$ precision; produces the Rudolphine Tables	Labelled training data with known noise level; dataset determines statistical power
Hypothesis class	Kepler proposes conic sections (circles, ovals, ellipses) as candidate orbit shapes	Model architecture: family of parametric curves from which the learner selects
Inductive bias	Parsimony: accept the simplest curve that fits the data to within observational noise	Regularisation: Occam's razor, MDL, AIC/BIC, sparsity penalty
Parameter estimation	Kepler performs ≈ 70 iterative orbital adjustments; converges to $a = 1.524\,\mathrm{AU}$, $e = 0.0934$ for Mars	Nonlinear least squares / Gauss–Newton; stochastic gradient descent for neural networks
Residual diagnostics	Kepler identifies 8' systematic error of circular model; fails to explain it as noise (4σ excess)	Validation loss; model misspecification tests; residual heteroscedasticity
Model expansion	Kepler abandons circles (null hypothesis rejected), adopts ellipse as minimal sufficient extension	Architecture search; adding hidden layers or parameters until validation loss stops improving
Compression / MDL	Ellipse = 2 parameters; Vicarious Hypothesis = 3 parameters; ellipse wins on parsimony and fit	Minimum description length; Kolmogorov complexity; bits to encode model and its residuals
Identification	Newton uses Kepler's three laws simultaneously to identify $F(r) = F_0 r^{-2}$ uniquely	Structural identification in causal models; using multiple experimental constraints to pin down a latent mechanism
Out-of-sample test	Moon test: inverse-square law at $r = 60.3\,R_\oplus$; agreement to 0.9%	Hold-out test set; domain generalisation benchmark
Transfer learning	Newton applies the same $F \propto r^{-2}$ to tides, comets, terrestrial gravity —without retraining	Zero-shot transfer across task domains; pre-trained foundation model applied without fine-tuning

Table 7: Comparison of Kepler, Newton, Bayesian inference, symbolic regression (SR), and deep learning (DL) on the planet-motion problem.

Feature	**Kepler (1609)**	**Newton (1687)**	**Bayesian / MCMC**	**SR / DL (AI)**
Optimisation	Iterative geometric trials (≈ 70 for Mars)	Geometric deduction from axioms	MCMC posterior sampling	Gradient descent (DL); tree search (SR)
Error metric	Maximum deviation ($> 8'$ threshold)	Logical consistency with all three Kepler laws	Log-likelihood; χ^2	MSE + complexity (MDL/Pareto)
Model search	Physical intuition (conic sections)	Inverse problem; unique solution	Prior over force laws	Tree search (SR); neural architecture (DL)
Noise handling	$8'$ threshold; accept if $< 2'$	Moon test residual 0.9%	Bayesian posterior integrates over noise	Neural smoothing + SR
Interpretability	Fully geometric; transparent	Fully deductive; transparent	Credible intervals; posterior	Expression recovered (SR); black-box (DL)
Universality	Five-planet induction	Moon test; deliberate proof	Marginal likelihood for model selection	Multi-system training data